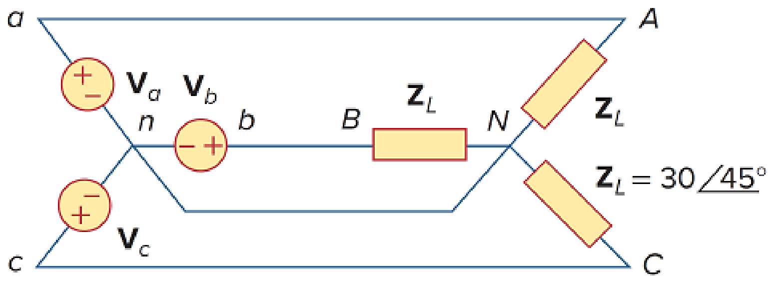
Chapter 12, Problem 30P

Problem

In Fig. 12.56, the rms value of the line voltage is 208 V. Find the average power delivered to the load.

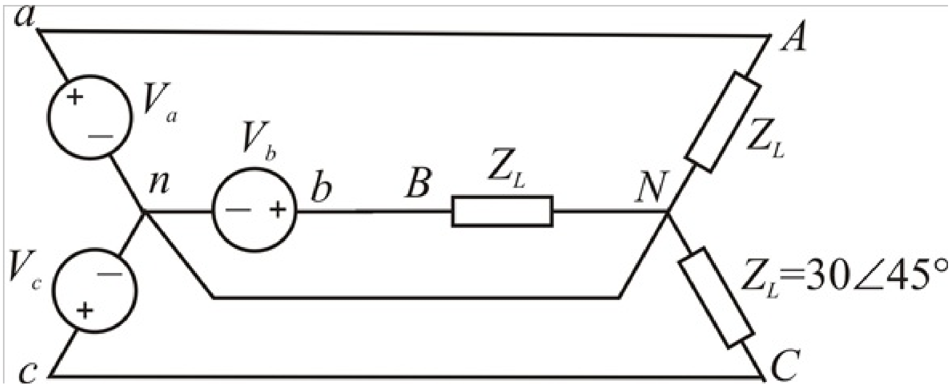
Figure 12.56



Step-by-step solution

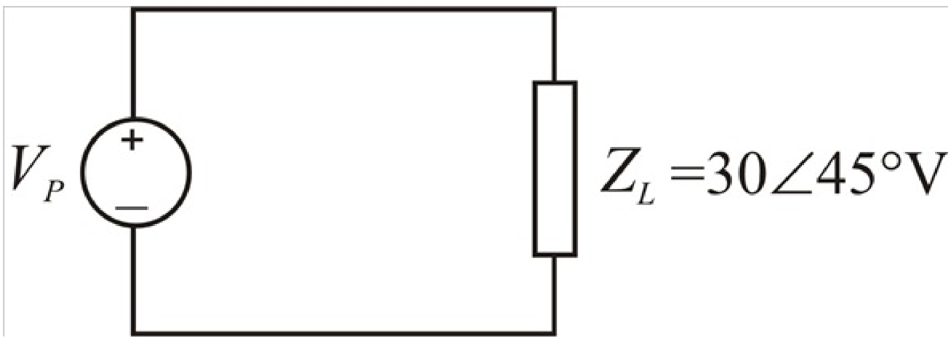
Step 1 of 3

Given circuit diagram



Step 2 of 3

Since this is a balanced system, we can take single phase equivalent circuit as shown below



Step 3 of 3

Given $V_L = 208\text{V}$ and $Z_L = 30\angle 45°\Omega$

Total complex power of the given circuit

$$S = 3S_p$$

$$S = \frac{3V_p^2}{Z_p^*}$$

We know that

$$V_P = \frac{V_L}{\sqrt{3}}$$

$$S = \frac{3\left(\frac{V_L}{\sqrt{3}}\right)^2}{Z_p^*}$$

$$S = \frac{V_L^2}{Z_p^*}$$

$$S = \frac{(208)^2}{30\angle 45°}$$

$$S = 1442.13\angle -45°\,\text{VA}$$

We know that $P = S\cos\theta$

$$P = 1442.13\cos(-45°)$$

$$P = 1442.13(0.7071)$$

$$P = 1019.74$$

$$\boxed{P = 1.02\text{kW}}$$